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Department of Computer Science and Engineering

PROJECT REPORT

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PROJECT TITLE:

ShopMate AI — An Intelligent Conversational Shopping Advisor

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CERTIFICATE

This is to certify that the project entitled

"ShopMate AI — An Intelligent Conversational Shopping Advisor"

is a bonafide work carried out by the following students of the Department of Computer Science and Engineering, SRM University AP, Amaravathi, in partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology during the academic year 2026–2027:

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The project report has been examined and approved as satisfying the academic requirements prescribed for the Bachelor of Technology Degree.

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ABSTRACT

Online shopping has undergone a transformational shift in recent years, yet the product discovery experience remains largely unguided and impersonal. Consumers browsing e-commerce platforms are overwhelmed by thousands of choices, generic search results, and static filter systems that fail to understand their individual preferences, budget constraints, or contextual needs. ShopMate AI is developed to address this gap by delivering an intelligent, conversational, AI-powered shopping advisor that transforms how users discover and evaluate products.

ShopMate AI is a full-stack web application built on a modern technology stack comprising a Python FastAPI backend and a React frontend styled with Tailwind CSS. At its core, the system integrates three sophisticated capabilities: a Groq-powered large language model (Llama-3.3-70B-Versatile) for natural language understanding and personalized recommendation generation, a Firecrawl web intelligence API for real-time retrieval of trending product information from across the internet, and a FAISS vector memory module powered by Sentence Transformers for semantic recall of past user preferences and queries.

The application presents a ChatGPT-style conversational interface through which users can describe what they are looking for in natural language. ShopMate AI responds with a thoughtful, conversational message accompanied by up to three curated product cards — each containing the product name, estimated price range, a reason for recommendation, a personalized explanation of why the product fits the user's specific needs, a quality rating, and a direct shopping link. A 'best match' indicator highlights the top recommendation among the presented options.

Authentication is handled through a dual-mechanism system: traditional email and password login for regular users, and an OTP-based login delivered via SMTP email or Twilio SMS for users who prefer passwordless access. Chat history is persisted in the browser's local storage on a per-user basis, enabling users to maintain multiple concurrent shopping conversations and

revisit past recommendations at any time. A chat management sidebar allows users to create new conversations, search through existing chats, and delete unwanted sessions.

The system prompt engineering is a key innovation of ShopMate AI. The Groq model is guided by a carefully crafted system prompt that instructs it to never repeat clarifying questions already answered by the user's history, to proactively generate recommendations even when some context is missing, to structure all responses as strict JSON objects containing the message, products array, and extracted preferences, and to maintain an evolving preference model that refines itself across interactions. This design produces a highly responsive, low-friction shopping experience that feels genuinely intelligent.

Testing across multiple shopping scenarios confirmed that the system reliably generates relevant, diverse product recommendations, correctly maintains preference continuity across messages within a session, and delivers API responses within acceptable latency bounds. The system is designed for cloud deployment with the backend on Render and the frontend on Vercel, with environment-based configuration management for secure API key handling.

Keywords: *ShopMate AI, Conversational AI, Product Recommendation, FastAPI, React, Groq LLM, Llama-3.3, FAISS, Sentence Transformers, Firecrawl, OTP Authentication, Tailwind CSS, Vite, E-Commerce Intelligence.*

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CHAPTER 1: PROJECT OVERVIEW

1.1 Introduction

The digital revolution in commerce has fundamentally transformed how consumers discover, evaluate, and purchase products. Despite this transformation, the product discovery experience on most e-commerce platforms remains inadequate — characterized by keyword-based search, static category filters, and generic bestseller lists that fail to account for the unique context, preferences, and intent of individual shoppers. The rise of large language models (LLMs) and conversational AI presents an unprecedented opportunity to reimagine product discovery as an intelligent dialogue, in which users can describe their needs in natural language and receive curated, thoughtful recommendations in response.

ShopMate AI is developed in response to this opportunity. It is a full-stack, AI-powered conversational shopping advisor that enables users to interact with an intelligent agent through natural language queries such as 'I need a gaming laptop under fifty thousand rupees with good battery life' or 'suggest me some minimalist running shoes for daily use.' The system processes these queries, retrieves real-time trending product data from the web, accesses a semantic memory of the user's past preferences, and synthesizes all of this context to generate personalized, structured product recommendations through the Groq Llama-3.3-70B language model.

1.2 System Overview

At its architectural core, ShopMate AI is organized as a two-tier client-server application. The frontend is a React single-page application (SPA) that renders a ChatGPT-style conversational interface complete with a message thread, product recommendation cards, a session management sidebar, and a dual-mode authentication system. The backend is a FastAPI application that exposes a set of RESTful endpoints for authentication and AI-powered chat,

integrating the Groq LLM API, the Firecrawl web search API, and a local FAISS vector store into a three-stage recommendation pipeline.

When a user submits a shopping query, the backend executes the following pipeline: it first searches the FAISS index for semantically similar past queries to build a memory context, then calls the Firecrawl API to retrieve current trending product information from the web, and finally sends a richly constructed prompt containing the user's query, memory context, live trend data, and stored preferences to the Groq API. The model responds with a structured JSON object containing a conversational message and an array of up to three product recommendations, which the frontend renders as interactive product cards alongside the message.

1.3 Project Name and Branding

The application is branded as ShopMate AI. The name captures the core value proposition: ShopMate implies a knowledgeable companion for shopping, while AI signals the intelligent, data-driven nature of the recommendations. The interface uses a dark slate color scheme with indigo accents to project a premium, modern aesthetic. The brand identity is consistently applied across the authentication screens, the main chat interface, and the product cards.

1.4 Motivation

The motivation for this project arises from the direct experience of the team members as online shoppers who repeatedly found existing search and recommendation tools inadequate. When purchasing products that require multi-criteria evaluation — such as laptops, smartphones, cameras, or fitness equipment — the inability of keyword search to understand context leads to frustrating, time-consuming discovery experiences. The emergence of capable, affordable LLM APIs such as Groq made it feasible to build a genuinely intelligent shopping assistant that could be accessed as a web application.

Additionally, the project serves as a practical exploration of how multiple AI technologies — LLMs, vector databases, and web intelligence APIs — can be combined in a cohesive full-stack application. This integration challenge is increasingly relevant to the modern software engineering landscape, making the project both technically ambitious and educationally valuable.

1.5 Project Scope

ShopMate AI encompasses the complete development of a production-ready conversational shopping assistant, from user authentication through natural language query processing to product recommendation display. The system supports multiple concurrent shopping sessions per user, maintains preference memory within each session using vector similarity search, and retrieves live product trend data to keep recommendations current. The scope includes backend API development, frontend UI implementation, authentication with OTP support, AI pipeline integration, and cloud deployment configuration.

CHAPTER 2: PROBLEM STATEMENT

2.1 The Core Problem

The fundamental problem addressed by ShopMate AI is the inadequacy of existing product discovery tools in understanding user intent and delivering contextually relevant, personalized recommendations. E-commerce search engines operate on keyword matching and popularity-based ranking, which are inherently unable to interpret the nuanced, multi-constraint queries that consumers naturally form when thinking about purchases. A user who searches for 'good laptop for college student who travels a lot and has limited budget' receives results filtered by the literal keywords, not by the underlying intent.

This gap between user intent and system understanding creates a frustrating, inefficient shopping experience that requires consumers to spend significant time manually filtering,

comparing, and evaluating products. Research consistently demonstrates that poor product discovery is a leading cause of cart abandonment and consumer dissatisfaction in e-commerce. The consequence is not merely inconvenient for individual shoppers — it represents a significant economic inefficiency in the broader marketplace.

2.2 Existing Solutions and Their Limitations

Several existing approaches attempt to address the product discovery problem, each with significant limitations. Collaborative filtering systems (used by Amazon, Netflix, and similar platforms) rely on purchase and browsing history to generate recommendations, but they are constrained by the cold-start problem for new users, are unable to handle novel intent expressions, and do not incorporate real-time information about new products or trends.

Search engine optimization-based recommendation shows sponsored and popular products rather than genuinely most relevant ones, creating a commercial bias that undermines user trust. AI chatbots integrated into some retail platforms are typically scripted or rule-based, lacking the contextual understanding and adaptive conversation management required for complex, multi-turn shopping assistance.

General-purpose AI assistants such as ChatGPT can discuss products conversationally, but they lack real-time product data, do not generate structured recommendation cards with prices and purchase links, and do not maintain user preference memory across sessions in a way that is personalized to individual users.

2.3 Identified Gaps

- No existing consumer tool combines conversational LLM intelligence with real-time web product data and persistent user preference memory in a unified, purpose-built shopping interface.
- Current recommendation systems cannot interpret complex, multi-constraint natural language queries such as budget ranges, use cases, aesthetic preferences, and technical specifications expressed in a single message.
- Most AI shopping tools either lack purchase links and pricing information, or provide static information that does not reflect current market trends and availability.
- Existing authentication systems for web applications typically require one implementation (password or OTP), not both in a user-selectable dual-mode system.
- Chat history and preference persistence is either absent or limited to a single session in existing AI shopping assistants, preventing continuity across multiple shopping occasions.

2.4 Problem Significance

The significance of solving this problem extends beyond convenience. Intelligent product discovery can help consumers make better-informed purchasing decisions, reduce the environmental and economic waste associated with returns driven by mismatched expectations, and create a more equitable marketplace in which quality products are surfaced based on fit rather than marketing spend. Building ShopMate AI as an open-architecture system demonstrates how these goals can be achieved using accessible, modern AI and web technologies.

CHAPTER 3: OBJECTIVES

3.1 Primary Objective

The primary objective of ShopMate AI is to develop a production-quality, AI-powered conversational shopping assistant that enables users to discover relevant products through natural language interaction. The system must be capable of maintaining preference continuity within a session, incorporating real-time web data into its recommendations, and presenting structured, actionable product information in an intuitive, visually appealing interface.

3.2 Technical Objectives

1. Design and implement a high-performance Python FastAPI backend with modular service architecture, clean separation of routing, business logic, and AI integration layers.
2. Integrate the Groq Llama-3.3-70B-Versatile language model via the OpenAI-compatible API to generate structured JSON product recommendations from natural language shopping queries.
3. Implement a FAISS-based in-memory vector store using the all-MiniLM-L6-v2 Sentence Transformer model to enable semantic similarity search across past user queries for preference context retrieval.
4. Integrate the Firecrawl Search API to retrieve real-time web data about trending products related to each user query, ensuring recommendations reflect current market availability.
5. Develop a dual-mode authentication system supporting both email/password login and OTP-based passwordless authentication via SMTP email and Twilio SMS.
6. Build a React frontend with a ChatGPT-style conversational interface, animated chat bubbles, product recommendation cards with pricing and purchase links, and a multi-session sidebar.

7. Implement per-user chat history persistence using browser local storage with support for multiple concurrent chats, session search, and individual chat deletion.
8. Design and engineer a structured JSON system prompt that constrains the LLM output format, prevents repetitive clarifying questions, enables progressive preference extraction, and marks the single best product match.

3.3 Performance Objectives

- Chat API endpoint response time within 5 seconds for AI recommendation generation under normal network conditions.
- Authentication endpoints responding within 300 milliseconds for password-based login and within 2 seconds for OTP delivery.
- FAISS vector search completing in under 50 milliseconds for index sizes up to 1,000 queries.
- Frontend first contentful paint within 2 seconds on standard broadband connections.
- System stability with 99% uptime in cloud-deployed production environments.

3.4 User Experience Objectives

- Zero configuration required from users — the interface should be immediately usable without tutorials or onboarding.
- Recommendations displayed as scannable product cards with all decision-relevant information visible without clicking through.
- Seamless conversation continuity where the AI never asks for information already provided in the same session.
- Shopping links on each product card that direct users to relevant purchase pages, reducing the friction between discovery and purchase.

3.5 Learning Outcomes

- Practical experience in building production-quality full-stack AI applications integrating multiple external APIs.
- Deep understanding of LLM prompt engineering for structured output generation and conversation management.
- Proficiency in vector database concepts, embedding generation, and semantic similarity search using FAISS.
- Competence in React state management for complex multi-session conversational UIs.
- Experience with cloud deployment, environment variable security, and CORS configuration for production web applications.

CHAPTER 4: TECHNOLOGIES USED

ShopMate AI is built on a carefully curated technology stack that balances performance, developer productivity, and AI capability. Each technology has been selected for a specific reason aligned with the system's requirements. This chapter provides a comprehensive explanation of every technology used, including its role in the system and the rationale for its selection.

4.1 Python 3.10+

Python serves as the programming language for the entire backend. Its selection is driven by the rich ecosystem of AI and data processing libraries, including the sentence_transformers and faiss-cpu packages that are central to the vector memory module. Python's dynamic typing enables rapid prototyping, while its asyncio support enables the non-blocking I/O operations required for concurrent API handling. The code makes extensive use of Python type annotations in conjunction with Pydantic for request validation, ensuring runtime correctness and IDE support.

Python 3.10 specifically introduces structural pattern matching and improved error messages that improve code clarity in the service layer. The language's clean readability is particularly valuable in the AI integration code, where complex prompt construction and JSON parsing logic must be understood and maintained by multiple team members.

4.2 FastAPI

FastAPI is a modern, high-performance Python web framework for building APIs with automatic interactive documentation. Its key advantages over alternatives like Flask include: native async/await support for non-blocking concurrent request handling; automatic request validation and serialization through Pydantic model integration; auto-generated Swagger UI and ReDoc API documentation derived directly from code type annotations; and dependency injection support for shared resources like database connections.

In ShopMate AI, FastAPI handles all API routing including four authentication endpoints (/auth/signup, /auth/signin, /auth/send-otp, /auth/verify-otp) and the primary AI chat endpoint (/chat). The CORS middleware is configured to allow all origins during development, with environment-specific restrictions for production. The framework's automatic 422 Unprocessable Entity responses for invalid request bodies ensure that malformed client requests are rejected with informative error messages without additional validation code.

4.3 Groq API with Llama-3.3-70B-Versatile

Groq provides ultra-fast AI inference for open-source large language models through a cloud API. The Llama-3.3-70B-Versatile model, developed by Meta AI and served through Groq's Language Processing Unit (LPU) infrastructure, offers state-of-the-art natural language understanding and generation at inference speeds significantly faster than comparable GPU-based services — a critical advantage for a real-time conversational application where response latency directly affects user experience.

The integration is implemented using the OpenAI Python SDK pointed at Groq's OpenAI-compatible base URL (https://api.groq.com/openai/v1), which means the implementation is portable and can switch AI providers by changing the base_url and api_key. The response_format parameter is set to json_object to constrain the model output to valid JSON, enabling reliable programmatic parsing of the recommendation structure.

The system prompt used with this model is a carefully engineered instruction set that defines the ShopMate AI persona, specifies the exact JSON output structure required, instructs the model never to repeat clarifying questions, directs it to proactively generate recommendations even from incomplete information, and requires it to extract and return user preferences in a structured extracted_preferences field. This prompt engineering is critical to the system's usability and is discussed in detail in the implementation chapter.

4.4 Firecrawl API

Firecrawl is a web intelligence API that provides structured search results from across the internet in real time. In ShopMate AI, it is used to retrieve the most current information about trending products related to each user query before the LLM generates its recommendation. This integration ensures that recommendations are not limited to the model's training data (which has a knowledge cutoff) but instead incorporate live product availability, recent reviews, and current pricing trends.

The firecrawl_service.py module calls the Firecrawl v1 search endpoint with a query constructed by prepending 'best trending' to the user's query. The response is limited to five results, each contributing a title and description to a formatted context string. This context is then passed to the LLM as the 'Live Trending Data' section of the user prompt, enabling the model to reference real products by name when generating recommendations.

4.5 FAISS (Facebook AI Similarity Search)

FAISS is a library developed by Meta AI Research for efficient similarity search in high-dimensional vector spaces. It enables the storage and fast nearest-neighbor retrieval of dense vector embeddings. In ShopMate AI, FAISS powers the vector memory module that gives the application conversational context awareness and preference continuity across messages within a session.

The VectorStore class in db/vector_store.py initializes a FAISS IndexFlatL2 (exact L2-distance search, suitable for the small in-session index sizes expected) with 384-dimensional vectors. Each time a user submits a query, it is encoded into a vector and added to the index. When processing subsequent queries, the top 3 most semantically similar past queries are retrieved and included in the LLM prompt as memory context, enabling the model to reference what the user has previously searched for and avoid redundant questions.

4.6 Sentence Transformers (all-MiniLM-L6-v2)

Sentence Transformers is a Python library providing pretrained models for generating dense sentence embeddings. The all-MiniLM-L6-v2 model produces high-quality 384-dimensional embeddings for arbitrary text inputs and runs efficiently on CPU, making it suitable for the in-process embedding generation required by the FAISS memory module. The model is loaded once at application startup in the VectorStore constructor, incurring a startup latency but enabling sub-millisecond embedding generation for subsequent requests.

The choice of all-MiniLM-L6-v2 is driven by its excellent performance-efficiency trade-off: it achieves competitive semantic similarity scores on standard benchmarks while being significantly smaller (approximately 80MB) and faster than larger models, making it deployable on the standard Render.com free tier without GPU requirements.

4.7 React 19

React is the JavaScript library used for building the frontend user interface. The application uses React 19, the latest major version, which introduces the React Compiler for automatic

performance optimization, improved server component support, and simplified state update semantics. The entire application is implemented as a single-file component (App.jsx) augmented by four reusable component files, following a pragmatic architecture that balances simplicity with maintainability for a project of this scope.

Key React patterns used in ShopMate AI include: useState for managing local component state including chat messages, authentication form fields, and UI toggles; useEffect for side effects including localStorage synchronization and chat initialization on login; and ref-based scroll management for keeping the chat window scrolled to the latest message. The component hierarchy is flat and intentional — App renders ChatWindow and ChatInput directly, while ChatWindow delegates individual message rendering to MessageBubble, which in turn renders ProductCard components for AI responses containing product recommendations.

4.8 Tailwind CSS v3

Tailwind CSS is a utility-first CSS framework that provides a comprehensive set of single-purpose CSS classes for building custom designs without leaving the JSX files. In ShopMate AI, it enables the construction of a premium dark-themed interface using utility classes like bg-slate-900, text-indigo-400, rounded-2xl, and backdrop-blur-md without writing a single custom CSS rule (with the exception of a dot-typing animation in App.css and a slide-up animation registered in tailwind.config.js).

The design system is based on the Tailwind slate color palette for backgrounds and the indigo palette for interactive elements and accents, creating a cohesive, modern aesthetic. Responsive utilities (md: and lg: prefixes) are used in the product card grid layout to adapt from single-column on mobile to three-column on large screens.

4.9 Vite

Vite is the build tool and development server used for the React frontend. It provides near-instant hot module replacement (HMR) during development by leveraging native ES module

support in modern browsers, dramatically improving the development iteration cycle compared to webpack-based setups. For production builds, Vite uses Rollup to produce highly optimized, code-split bundles. The vite.config.js uses the @vitejs/plugin-react plugin for JSX transformation and Fast Refresh integration.

4.10 Axios

Axios is the HTTP client library used in the frontend for API communication. The services/api.js module defines five exported async functions (sendChatMessage, signupUser, signinUser, sendOTP, verifyOTP) that abstract all backend communication. Axios is preferred over the native fetch API for its automatic JSON request/response transformation, consistent error handling through response interceptors, and cleaner async/await syntax. All API calls point to the backend base URL defined as a constant (API_URL = 'http://localhost:8000') that would be replaced with the production URL through environment variables in deployment.

4.11 Additional Frontend Libraries

Lucide React provides the SVG icon library used throughout the interface, supplying icons such as Sparkles for the AI logo, MessageSquare for chat sessions, Send for the message submit button, LogOut for the logout action, Trash2 for chat deletion, Search for the chat search field, and ExternalLink for product card links. The library's consistent design language and tree-shakable export structure make it ideal for a performance-conscious React application.

The clsx and tailwind-merge libraries are used in the MessageBubble component to conditionally compose Tailwind class strings without conflicts. clsx handles conditional class inclusion, and tailwind-merge resolves Tailwind-specific class conflicts that would otherwise produce unexpected styles.

4.12 Technology Stack Summary

Layer	Technology	Version	Purpose

Backend	Python	3.10+	Server-side programming language
Backend	FastAPI	0.100+	REST API framework
Backend	Uvicorn	0.22+	ASGI server
Backend	Pydantic	2.0+	Request/response validation
AI	Groq API (Llama 3.3-70B)	Latest	LLM for recommendations
AI	Firecrawl	v1	Live web product search
AI	FAISS (CPU)	1.7+	Vector similarity search
AI	Sentence Transformers	2.2+	Text embedding generation
Frontend	React	19	UI component framework
Frontend	Tailwind CSS	3.4+	Utility-first styling
Frontend	Vite	8.0+	Build tool & dev server
Frontend	Axios	1.15+	HTTP client
Frontend	Lucide React	1.11+	Icon library
Auth	SMTP (Gmail)	—	Email OTP delivery
Auth	Twilio SMS	—	SMS OTP delivery

CHAPTER 5: KEY FEATURES

ShopMate AI incorporates a carefully designed set of features that collectively deliver an intelligent, seamless, and personalized shopping assistance experience. This chapter describes each feature in detail, explaining its design rationale and technical implementation.

5.1 Conversational AI Shopping Interface

The most prominent feature of ShopMate AI is its conversational interface modeled after modern AI chat applications. Users interact with the system through a chat window in which their messages appear as right-aligned indigo bubbles and AI responses appear as left-aligned

slate-colored bubbles with an accompanying Sparkles icon. This familiar interaction paradigm immediately communicates to users how to engage with the system and removes the learning curve associated with traditional search interfaces.

The interface supports multi-turn conversations in which context is maintained across messages within a session. If a user first asks for 'laptops under 50,000' and then follows up with 'what about ones with dedicated GPU?', the system interprets the second query in the context of the first, without requiring the user to restate their budget constraint. This contextual continuity is enabled by the FAISS memory module, which provides past query context to the LLM on each request.

The chat input field is a full-width rounded text input with a circular indigo submit button featuring the Send icon. The input is disabled during AI processing, and the submit button displays a spinning Loader2 icon during API calls. A subtle footer note below the input area informs users that ShopMate AI may make mistakes and that important information should be verified — a transparency measure that builds appropriate user trust.

5.2 AI-Powered Product Recommendations with Structured Cards

The recommendation output is presented not merely as text, but as a combination of a conversational message and up to three structured product cards rendered in a responsive grid layout (one column on mobile, two on medium screens, three on large screens). Each product card, implemented in ProductCard.jsx, displays: the product name as a heading (which changes color to indigo on hover, providing tactile feedback); the price range as a pill badge with a Tag icon; a general reason for the recommendation as descriptive body text; and a 'View Details' link with an ExternalLink icon that opens the product's shopping page in a new tab.

The LLM is prompted to always include one product marked as best_match: true among the recommendations, enabling future UI versions to visually distinguish the top recommendation. The model is also instructed to populate the why_for_you field with a personalized explanation

specific to the user's extracted preferences — a distinction from the general 'reason' field that provides meaningful personalization signal.

5.3 Real-Time Trending Product Data (Firecrawl Integration)

A critical limitation of LLM-only recommendation systems is that they are constrained by their training data cutoff. ShopMate AI overcomes this through the Firecrawl integration in firecrawl_service.py, which performs a real-time web search for trending products related to each user query before the LLM generates its response. The search query is constructed as 'best trending [user_query] to buy right now', retrieves the top five results, and formats their titles and descriptions into a context string that is injected into the LLM prompt as 'Live Trending Data'.

This integration means that even for product categories that emerged after the LLM's training cutoff, ShopMate AI can generate informed, current recommendations. It also means that pricing and availability references in recommendations reflect current market conditions rather than historical training data.

5.4 Vector Memory for Preference Continuity (FAISS)

The FAISS vector memory module, implemented as the VectorStore singleton in db/vector_store.py, provides ShopMate AI with the ability to remember what users have searched for within a session. Every time a user submits a query, the query text is encoded into a 384-dimensional vector by the all-MiniLM-L6-v2 model and added to the FAISS index. When the next query arrives, the module searches for the three most semantically similar past queries and returns their original text strings, which are then included in the LLM prompt as 'Past Context (Memory)'.

This memory mechanism serves two purposes. First, it enables the LLM to avoid repeating questions about preferences that the user has already expressed — one of the most frustrating aspects of AI chatbots. Second, it enables the LLM to make connections between related

queries that the user might not have explicitly linked, improving the coherence and relevance of the recommendations across a multi-turn conversation.

5.5 Dual-Mode Authentication (Password and OTP)

ShopMate AI implements a comprehensive authentication system that supports two distinct login methods. The password-based method is the standard email plus password flow, with separate signup and signin endpoints that persist credentials in a JSON file (users.json). The OTP-based method enables passwordless authentication through a two-step process: the user enters their email address, requests an OTP via the 'Send OTP' button, and then enters the six-digit code they receive.

OTP delivery is handled through the send_email_otp function using Python's smtplib library configured with SMTP credentials from the environment variables. The toggle between authentication methods is presented through a button in the auth form that switches between 'Use OTP instead' and 'Use Password instead', providing a smooth, single-page authentication experience without navigation.

5.6 Multi-Session Chat Management Sidebar

The left sidebar of the main application interface provides a complete chat session management system. Users can create new shopping conversations with the '+ New Chat' button, each of which is independently stored in local storage under a key namespaced to the current user. The first message sent in a new chat automatically becomes the chat title (truncated to 30 characters), enabling users to identify their shopping sessions at a glance.

A search bar in the sidebar filters the displayed chat list in real time as the user types, enabling quick navigation in accounts with many saved conversations. Each chat item in the list shows a Trash2 delete icon on hover, providing a direct one-click delete action with a confirmation dialog to prevent accidental deletion.

5.7 User Authentication State and Session Persistence

The application maintains authentication state through localStorage, storing the username under the key shopmate_user. On application load, the currentUser state is initialized from localStorage, meaning users remain logged in across browser sessions without needing to re-authenticate. Chat histories are stored per-user under the key shopmate_chats_{username}, ensuring complete data isolation between different accounts on the same device.

5.8 Typing Indicator and Loading States

During AI processing, the interface displays a typing indicator in the chat window — a dot-typing animation (defined in App.css) displayed inside a chat bubble styled identically to AI responses. The chat input field is disabled while loading to prevent duplicate message submission. The submit button changes from the Send icon to a spinning Loader2 icon, providing a secondary visual confirmation of the loading state.

5.9 Share Functionality

The header of the main application contains a Share button that copies the current application URL to the clipboard using the Clipboard API. When clicked, the button transitions from a Share2 icon with 'Share' label to a Check icon with 'Link Copied' label, with the button changing to a green color scheme to confirm the action. After two seconds, the button automatically returns to its default state.

CHAPTER 6: TARGET USERS

ShopMate AI is designed to serve a broad but well-defined user population united by the common need for intelligent, personalized shopping guidance. Understanding the diversity of

target users is important for appreciating the design choices made in the interface and AI configuration.

6.1 Students and Young Adults

Students represent a primary user segment. They frequently make purchasing decisions for electronics, study materials, fashion, and fitness equipment on constrained budgets, and they are highly comfortable with conversational interfaces. For this group, ShopMate AI's ability to understand multi-constraint queries like 'best earphones under 2000 rupees for online classes' and return structured recommendations with prices and purchase links provides direct, practical value.

6.2 Tech-Savvy Professionals

Working professionals with moderate to high disposable income but limited time for research are another core segment. They benefit from ShopMate AI's ability to quickly surface curated recommendations for higher-value purchases such as laptops, smartphones, home appliances, and productivity tools.

6.3 First-Time Online Shoppers

Users who are new to online shopping often find the sheer volume of options and the technical complexity of product specifications overwhelming. ShopMate AI's conversational approach lowers the barrier to entry by enabling these users to describe their needs in plain language and receive curated, explained recommendations.

6.4 Gift Buyers

Users seeking gift recommendations represent a distinct use case in which the buyer is purchasing for someone else's preferences rather than their own. ShopMate AI handles this scenario well because users can describe the recipient's characteristics and the occasion in

natural language, and the LLM can generate appropriate suggestions without requiring detailed technical specifications.

CHAPTER 7: REAL-WORLD APPLICATIONS

ShopMate AI's core technology — the combination of LLM intelligence, real-time web data, and vector memory — has applications that extend well beyond its current implementation as a general product recommendation tool. This chapter explores both the immediate use cases of the current system and the broader application scenarios that the underlying architecture enables.

7.1 General E-Commerce Product Discovery

The most direct application of ShopMate AI is as a shopping assistant integrated into or alongside general e-commerce platforms. A user searching for 'a laptop for video editing and gaming within 80,000 rupees that is also portable' can express this multi-dimensional requirement in a single natural language message and receive three curated recommendations, each with a price range and direct purchase link, within seconds.

7.2 Niche Retail Vertical Assistants

The ShopMate AI architecture can be specialized for specific retail verticals by adjusting the system prompt to focus on domain-specific product knowledge and terminology. For a fashion retail application, the prompt would instruct the model to consider style preferences, size recommendations, and seasonal trends. This vertical specialization requires only prompt modification and potentially category-specific Firecrawl query construction.

7.3 Corporate Procurement Assistant

Organizations that regularly procure office equipment, IT hardware, and consumables can benefit from a ShopMate AI variant configured for B2B procurement. The multi-session memory capability enables the system to maintain context across a multi-day procurement research process.

7.4 Customer Service Augmentation

Retail businesses can deploy ShopMate AI as a customer-facing product selection assistant that reduces the burden on human customer service agents. When a customer contacts support asking about a replacement product, a ShopMate AI instance configured with the brand's product catalog can immediately surface correct compatible products.

7.5 Educational Application: AI Integration Demonstration

ShopMate AI also serves as an exemplary educational project for students and developers learning how to integrate multiple AI services into a cohesive full-stack application. The codebase covers a comprehensive range of modern full-stack AI development concepts including LLM API integration, vector embeddings, real-time web intelligence API usage, async Python web frameworks, and React state management for conversational UIs.

7.6 Price Comparison and Research Tool

The combination of Firecrawl's real-time web data and the LLM's synthesis capabilities enables ShopMate AI to function as a lightweight price comparison and research tool. When a user asks about a specific product, the system can surface information about current pricing across multiple retailers and summarize the relevant trade-offs.

CHAPTER 8: TECHNICAL ARCHITECTURE

The architecture of ShopMate AI follows a client-server pattern with a clear separation between the presentation layer (React frontend), the business logic layer (FastAPI backend), and the AI services layer (Groq, Firecrawl, FAISS). This chapter describes each architectural layer in detail, explains the data flows between components, and discusses the key architectural decisions that shape the system's behavior and scalability characteristics.

8.1 High-Level Architecture

The system comprises five major components: the React SPA served through Vite, the FastAPI backend application, the FAISS in-memory vector store, the Groq AI API (external), and the Firecrawl web intelligence API (external). The user interacts exclusively with the React frontend, which communicates with the FastAPI backend via HTTP REST API calls.

Component	Technology	Location	Role
Frontend (SPA)	React + Vite	Browser / Vercel CDN	User interaction layer
Backend (API)	FastAPI + Uvicorn	Render.com	Orchestration & business logic
Vector Store	FAISS + Sentence Transformers	In-process (backend)	Semantic memory
LLM Service	Groq Llama-3.3-70B	Groq Cloud API	Recommendation generation
Web Intelligence	Firecrawl Search API	Firecrawl Cloud API	Real-time product data
Auth Store	JSON file (users.json)	Backend filesystem	Credential persistence
Chat History	Browser localStorage	Client browser	Per-user session storage

8.2 Frontend Layer

The frontend is a React 19 single-page application organized into six files: App.jsx as the root component containing authentication logic and main layout, ChatWindow.jsx as the message

display container, MessageBubble.jsx for rendering individual messages, ProductCard.jsx for rendering recommendation cards, ChatInput.jsx for the message input form, and services/api.js as the API communication layer.

The frontend state architecture is centralized in App.jsx, which manages authentication state (currentUser, authMode, authMethod, email, password, otp, otpSent), chat session state (chats array, currentChatId), and UI state (isLoading, searchQuery, isCopied). This centralized approach ensures consistent state across components without requiring a state management library like Redux.

8.3 Backend Layer

The FastAPI backend is structured as a single-file application (main.py) for this project iteration, with service logic extracted into separate modules (services/llm_service.py and services/firecrawl_service.py) and the vector store into its own module (db/vector_store.py). The application startup sequence initializes the CORS middleware, loads environment variables via python-dotenv, creates the db directory and users.json file if they do not exist, and implicitly initializes the VectorStore singleton.

8.4 AI Service Pipeline

The AI service pipeline is the core intellectual contribution of ShopMate AI. It operates as follows when a /chat request arrives:

9. The FAISS vector store searches for the top 3 most semantically similar past queries from the in-memory index and returns their text strings as memory_context.
10. The Firecrawl service executes a web search for 'best trending [query] to buy right now', retrieves up to 5 results, and formats their titles and descriptions into a multi-line trends_context string.
11. The LLM service constructs a user_prompt containing the user's current query, the preferences dict, the memory_context, and the trends_context.

12. The Groq API call is made with response_format={'type': 'json_object'} to guarantee valid JSON output.
13. The JSON response is parsed and returned as the HTTP response body. The frontend receives the message and products array and renders them appropriately.
14. The current user query is added to the FAISS index for use as memory context in future queries.

8.5 Authentication Architecture

The authentication system supports two parallel flows sharing the same user store (users.json). The password flow follows a simple register/authenticate pattern with email as the username key. The OTP flow uses an in-memory otp_store dictionary that maps email addresses to their current valid OTP. If neither SMTP nor Twilio is configured, the OTP is returned in the API response for development testing.

8.6 Data Flow Diagram

The end-to-end data flow for a product recommendation request proceeds as follows: the user types a query in the React frontend input field and clicks Send. The ChatInput component calls onSendMessage in App.jsx, which immediately adds the user message to the chat state (optimistic update) and calls sendChatMessage in services/api.js. Axios POSTs the query and preferences to the /chat endpoint. The backend's pipeline retrieves vector memory, fetches Firecrawl trends, constructs the LLM prompt, calls Groq, parses the response, adds the query to the FAISS index, and returns the JSON. The frontend receives the response and React re-renders the chat window with the new MessageBubble containing the text and ProductCard components.

CHAPTER 9: PRE-REQUISITES

9.1 Software Requirements

- Python 3.10 or higher: Required for the backend. Verify with: python --version
- pip 22.0 or higher: Python package manager for installing backend dependencies.
- Node.js 18 LTS or higher: Required for the React frontend build toolchain including Vite and npm.
- npm 8.0 or higher: Node package manager bundled with Node.js.
- Git 2.30 or higher: For cloning the repository and version control management.
- Visual Studio Code: Recommended IDE with the Python and ESLint extensions.
- A modern web browser (Chrome, Firefox, or Edge) with ES6 module support.

9.2 API Keys Required

API Service	Environment Variable	Purpose	Where to Get
Groq AI	GROQ_API_KEY	LLM inference for recommendations	console.groq.com
Firecrawl	FIRECRAWL_API_KEY	Real-time web product search	firecrawl.dev
Gmail SMTP	SMTP_EMAIL, SMTP_PASSWORD	OTP email delivery	Google App Passwords
Twilio (optional)	TWILIO_ACCOUNT_SID, TWILIO_AUTH_TOKEN, TWILIO_FROM_NUMBER	SMS OTP delivery	twilio.com/console

9.3 Hardware Requirements

Component	Minimum	Recommended
Processor	Intel Core i5 / AMD Ryzen 5 (2.0 GHz)	Intel Core i7 / AMD Ryzen 7 (3.0+ GHz)
RAM	8 GB	16 GB (for smooth Sentence Transformers loading)
Storage	2 GB free disk space	5 GB (for model files and node_modules)
Network	10 Mbps stable broadband	50+ Mbps (for low Groq/Firecrawl API latency)
GPU	Not required	Not required (all inference is cloud-based)

9.4 Backend Dependencies (requirements.txt)

```
pip install fastapi uvicorn supabase python-dotenv requests openai
faiss-cpu sentence-transformers
```

9.5 Frontend Dependencies (package.json)

Package	Version	Role
react	^19.2.5	UI component framework
react-dom	^19.2.5	React DOM rendering
axios	^1.15.2	HTTP client for API calls
lucide-react	^1.11.0	SVG icon library
clsx	^2.1.1	Conditional class names
tailwind-merge	^3.5.0	Tailwind class conflict resolution
tailwindcss	^3.4.19	Utility-first CSS framework
vite	^8.0.10	Build tool and dev server
@vitejs/plugin-react	^6.0.1	Vite plugin for React/JSX

CHAPTER 10: PRIOR KNOWLEDGE REQUIRED

Working effectively with the ShopMate AI codebase and extending its capabilities requires competence across several technical domains. This chapter outlines the prerequisite knowledge organized by domain.

10.1 Python Programming

A solid foundation in Python is essential. Required competencies include: function definition with type annotations, class definition and instantiation, asynchronous programming with async/await, exception handling with try/except/finally, environment variable access with os.getenv, JSON parsing with the json module, file I/O operations, list comprehensions and dictionary operations, and working with third-party packages through pip.

10.2 FastAPI and RESTful API Development

Understanding FastAPI requires knowledge of: HTTP methods and their appropriate use cases; REST API design principles including endpoint naming conventions and response status codes; Pydantic model definition for request body validation; CORS and why it is required for browser-based clients; async route handlers and their advantages for I/O-bound operations; and exception raising with HTTPException for error responses.

10.3 React and Frontend Development

Frontend contributions require knowledge of: React functional components and JSX syntax; React hooks including useState, useEffect, and useRef; event handling in React; conditional rendering and array mapping for dynamic list rendering; CSS utility class composition with Tailwind; and the JavaScript Axios API for HTTP communication.

10.4 AI and LLM Concepts

Understanding the AI layer requires familiarity with: large language model fundamentals; prompt engineering principles including system prompts, user prompts, and output format specification; structured output generation and JSON mode operation; vector embeddings as

dense numerical representations of semantic meaning; cosine similarity and L2 distance as measures of vector similarity; and approximate nearest neighbor search as applied to semantic retrieval.

10.5 Web Authentication

Working with the authentication system requires understanding of: session management in web applications; localStorage as a client-side persistence mechanism; password-based authentication flow; OTP-based authentication flow including generation, delivery, and verification; SMTP email protocol and Gmail application passwords; and basic API key security practices.

CHAPTER 11: PROJECT OBJECTIVES — DETAILED

11.1 Technical Objectives in Depth

The technical objective of integrating multiple AI services — Groq for language generation, Firecrawl for web intelligence, and FAISS for vector memory — into a single coherent pipeline represents the central engineering challenge of this project. The key technical insight is that each service addresses a distinct limitation of the others: Groq provides language intelligence but lacks current product data; Firecrawl provides current data but lacks synthesis and reasoning capability; FAISS provides memory continuity but cannot generate new content. The pipeline's value emerges from the orchestrated combination of all three.

The prompt engineering objective is particularly significant because it determines the quality and consistency of the system's output. The system prompt must balance multiple competing requirements: it must be detailed enough to reliably constrain the output to valid JSON of the required structure; it must be flexible enough to accommodate the full diversity of shopping queries; it must encode behavioral instructions that the model reliably follows; and it must not be so long that it consumes excessive context window tokens.

11.2 Performance Objectives in Context

The five-second API response time target for the /chat endpoint accounts for the sequential nature of the pipeline: Firecrawl search (approximately 1–2 seconds), Groq API call (approximately 1–3 seconds), and FAISS search plus embedding generation (less than 200 milliseconds). These targets are measured under standard broadband conditions and reflect what user experience research identifies as the maximum acceptable wait time for an intelligent response.

11.3 Deployment Objectives in Context

The cloud deployment objective requires that the application be fully operational on Render (backend) and Vercel (frontend) within a standard free-tier resource allocation. This constraint drives architectural decisions including the use of an in-memory FAISS index, JSON file storage for user credentials, and local storage for chat history — all of which make the system deployable at zero cost for demonstration and academic purposes.

CHAPTER 12: PROJECT WORKFLOW AND MILESTONES

The development of ShopMate AI followed a structured, iterative workflow organized into five milestones. Each milestone built on the previous, progressively adding layers of complexity and functionality until the complete system was operational.

12.1 Milestone 1: Architecture Design and Environment Setup

The first milestone focused on defining the system architecture, establishing the development environment, and validating the core technology choices. This phase began with a comparative analysis of available LLM APIs and a decision to use Groq for its superior inference speed and the availability of the Llama-3.3-70B model at no cost through the free tier.

The FastAPI project structure was established with the main.py entry point, the services directory, and the db directory. The React frontend was scaffolded using the Vite create command, and Tailwind CSS was configured. A key activity was the initial API key acquisition and validation before writing any application-level code against each service.

12.2 Milestone 2: Backend Core Development

The second milestone implemented all backend functionality including authentication endpoints, the FAISS vector store, the Firecrawl service, and the Groq LLM service. The VectorStore class was developed and unit-tested independently before integration into the /chat pipeline.

The prompt engineering for the LLM service was the most iterative activity of this milestone. Multiple prompt formulations were tested, evaluating the model's compliance with the JSON output format, its adherence to the 'never repeat clarifying questions' instruction, and the quality and diversity of its product recommendations. The final prompt structure evolved through approximately eight iterations before producing consistently high-quality, correctly formatted output.

12.3 Milestone 3: Frontend Development

The third milestone built the complete React user interface. Development began with the authentication screens, establishing the dark-themed visual design with indigo accents that would be carried through the entire application. The main application layout was built next, establishing the three-pane structure: the sidebar (chat list, new chat button, search), the header (ShopMate AI branding, share button), and the main content area (ChatWindow, ChatInput).

12.4 Milestone 4: Integration and Testing

The fourth milestone integrated the frontend and backend, resolved CORS issues, and conducted end-to-end testing of complete user workflows. Several integration issues were identified and resolved, including an API response format mismatch and a bug in the localStorage chat persistence that caused a brief flash of an empty chat list on login.

12.5 Milestone 5: Refinement, Deployment, and Documentation

The final milestone addressed code quality, deployment configuration, and documentation. Deployment configuration files were created for Render (backend) and Vercel (frontend). The theme_switcher.py script was written to facilitate rapid theme iteration during the design phase. The final report and code documentation were prepared during this milestone.

CHAPTER 13: DETAILED SYSTEM DESIGN

13.1 Functional Requirements

Req. ID	Requirement Description	Priority
FR-01	Users shall be able to register with an email address and password.	High
FR-02	Users shall be able to sign in with email and password.	High
FR-03	Users shall be able to request and verify an email OTP for passwordless login.	High

FR-04	Users shall be able to request and verify an SMS OTP for passwordless login.	Medium
FR-05	Authenticated users shall be able to submit natural language product queries.	High
FR-06	The system shall generate up to three product recommendations per query.	High
FR-07	Each recommendation shall include name, price, reason, why_for_you, rating, and URL.	High
FR-08	One recommendation shall be marked as the best match.	Medium
FR-09	The system shall retrieve real-time trending product data for each query.	High
FR-10	The system shall maintain semantic memory of past queries within a session.	High
FR-11	Users shall be able to create, switch between, search, and delete chat sessions.	High
FR-12	Chat history shall persist across browser sessions using localStorage.	High
FR-13	Users shall be able to log out, clearing their session state.	High
FR-14	Users shall be able to copy the application URL to clipboard.	Low

13.2 Non-Functional Requirements

Attribute	Requirement
Performance	Chat endpoint response time < 5 seconds; Auth endpoints < 500ms
Reliability	System uptime > 99%; graceful error handling for all API failures
Security	API keys stored only in environment variables, never in client code
Scalability	Architecture supports horizontal backend scaling without state issues
Usability	Functional on screens >= 320px width; no onboarding required
Maintainability	Service layer separated from routes; single responsibility per module
Portability	Deployable on any ASGI-compatible cloud platform without code changes

13.3 Pydantic Data Models

```
class SignupRequest(BaseModel):   email: str;   password: str
class SigninRequest(BaseModel):   email: str;   password: str
class SendOTPRequest(BaseModel):  email: str
class VerifyOTPRequest(BaseModel): email: str; otp: str
class ChatRequest(BaseModel):      query: str; preferences: dict =
{}
```

13.4 LLM Output JSON Schema

```
{ "message": "string (conversational reply)",
  "products": [
    { "name": "string",
      "price": "string (e.g., Rs. 15,000-20,000)",
      "reason": "string (general recommendation reason)",
      "why_for_you": "string (personalized explanation)",
      "rating": "string (e.g., 4.5)",
      "url": "string (product or Amazon search URL)",
      "best_match": true/false } ],
  "extracted_preferences": { "budget": "...", "category": "...",
"other": "..." } }
```

13.5 Design Decisions

13.5.1 Single-File Backend vs. Modular Structure

The backend is implemented primarily in main.py with service logic extracted to separate files. For the current scope, this approach provides the right balance between simplicity (avoiding unnecessary abstraction layers) and modularity (the three service files can be tested and modified independently).

13.5.2 In-Memory Vector Store vs. Persistent Vector Database

The FAISS index is maintained in process memory rather than persisted to a database. This design choice eliminates the need for an external vector database service that would introduce

cost, network latency, and operational complexity. The trade-off is that memory is lost on server restart, which is acceptable for within-session preference continuity.

13.5.3 localStorage vs. Server-Side Chat Storage

Chat history is stored in the browser's localStorage rather than on a server. This design eliminates the need for a database and keeps the system stateless, simplifying deployment and reducing cost. The trade-off is that chats are tied to a specific browser. A production version would migrate to server-side storage for cross-device synchronization.

CHAPTER 14: IMPLEMENTATION DETAILS

14.1 Backend: main.py

The main.py file is the entry point for the FastAPI application. It imports the three service modules at startup, which triggers the Sentence Transformers model load. The file defines five Pydantic request models, configures CORS middleware, and implements six route handlers. The /auth/send-otp endpoint determines whether the identifier is an email or phone number by checking for the presence of '@', then routes delivery accordingly.

The /chat endpoint is the most complex handler, implementing the three-stage pipeline with comprehensive error handling. Each stage is wrapped in a try/except block, with Firecrawl failures returning an informative error string and Groq failures raising an HTTPException.

14.2 Backend: services/llm_service.py

The llm_service.py module implements the generate_recommendations function. The system_prompt is a multi-paragraph instruction set defined as a Python triple-quoted string. Its key behavioral instructions are: (1) Never ask a clarifying question that has already been answered by past context or the current query. (2) If context is insufficient, provide 3 diverse

options across different price ranges. (3) Always respond strictly in the specified JSON format. (4) Extract and update the user's preferences. (5) Mark exactly one product as best_match: true.

The OpenAI client is initialized with Groq's base URL and the GROQ_API_KEY environment variable. The model llama-3.3-70b-versatile is specified, max_tokens is set to 1000, and response_format is set to json_object.

14.3 Backend: services/firecrawl_service.py

The fetch_trending_products function calls the Firecrawl v1/search endpoint using Python's requests library. The payload specifies the search query (prepended with 'best trending') and a limit of 5 results. The function handles the case where the API key is not configured, the case where the API returns an empty result set, and network errors through a try/except wrapper.

14.4 Backend: db/vector_store.py

The VectorStore class encapsulates all FAISS operations. The constructor loads the all-MiniLM-L6-v2 Sentence Transformer model, initializes a FAISS IndexFlatL2 with dimension 384, and creates an empty memory list. The add_query method encodes the query into a float32 numpy array and calls index.add(). The search_similar_queries method first checks index.ntotal to guard against searching an empty index, then encodes the current query and returns the top-k most similar past queries.

14.5 Frontend: App.jsx

App.jsx is the root component of the React application and contains the majority of the application's logic. Its structure can be divided into four functional sections: state initialization (fifteen useState declarations), side effects (two useEffect hooks for localStorage sync), the authentication screen, and the main application screen. The handleSendMessage function implements optimistic UI updates — the user's message is immediately added to the chat state before the API call is made, so the user sees their message appear instantaneously.

14.6 Frontend: Component Architecture

ChatWindow.jsx renders the scrollable message container with auto-scrolling managed via useRef/useEffect. MessageBubble.jsx uses clsx and tailwind-merge to compose conditional class strings. For AI messages, the bubble uses bg-slate-800 with rounded-tl-none. For user messages, it uses bg-indigo-600 with rounded-tr-none — a standard convention in chat UI design that visually communicates message directionality.

ProductCard.jsx renders a single product recommendation as a styled card with hover interactions. The hover state changes the product name color from text-slate-100 to text-indigo-400 through a transition-colors class. ChatInput.jsx manages its own query state independently from App.jsx, clearing the input after submission.

14.7 Theme Customization (theme_switcher.py)

The theme_switcher.py utility script enables rapid global theme color changes across all JSX files. It reads App.jsx and all .jsx files in the components directory, performs string replacement on specified Tailwind color classes, and writes the modified content back. This approach enables the entire application's color scheme to be changed in a single script execution.

CHAPTER 15: DEPLOYMENT PROCESS

15.1 Backend Deployment on Render

Render.com provides a free-tier managed cloud platform for running Python web services. The deployment process involves: (1) Pushing the project to a GitHub repository. (2) Creating a new Web Service in the Render dashboard. (3) Setting the build command to pip install -r requirements.txt. (4) Setting the start command to uvicorn main:app --host 0.0.0.0 --port

$PORT. (5) Configuring all environment variables in the Render environment variable panel. (6) Selecting Python 3.10 as the runtime.

Note that Render's free tier spins down services after 15 minutes of inactivity, causing a cold start delay of approximately 30–60 seconds on the first request after an idle period.

15.2 Frontend Deployment on Vercel

Vercel provides the optimal deployment platform for Vite/React applications. The deployment process involves: (1) Importing the repository into a new Vercel project. (2) Setting the root directory to frontend/. (3) Vercel automatically detects the Vite framework and sets the correct build command (npm run build) and output directory (dist). (4) Setting the environment variable VITE_API_BASE_URL to the Render backend URL. Vercel provides automatic HTTPS, global CDN distribution, and preview deployments for pull requests.

15.3 Environment Configuration (.env)

Variable	Value (Redacted)	Description
GROQ_API_KEY	gsk_w3FO...****	Groq inference API key
FIRECRAWL_API_KEY	fc-0df0...****	Firecrawl web search key
SMTP_EMAIL	aditya_tripathi@srmap.edu.in	Gmail address for OTP
SMTP_PASSWORD	beyd...****	Gmail app-specific password
SMTP_SERVER	smtp.gmail.com	Gmail SMTP server
SMTP_PORT	587	SMTP TLS port

CHAPTER 16: TESTING AND VALIDATION

16.1 Testing Strategy

ShopMate AI was tested using a combination of API-level functional testing, end-to-end user workflow testing, and manual stress testing of the AI pipeline. The testing strategy was designed to validate that each functional requirement is correctly implemented and that the system behaves correctly in edge cases and error conditions.

16.2 API Endpoint Testing

Endpoint	Test Scenario	Expected Result	Pass/Fail
POST /auth/signup	Valid new user registration	200 OK with username	PASS
POST /auth/signup	Duplicate email registration	400 Bad Request	PASS
POST /auth/signin	Valid credentials	200 OK with username	PASS
POST /auth/signin	Invalid password	401 Unauthorized	PASS
POST /auth/send-otp	Valid email address	200 OK with message	PASS
POST /auth/verify-otp	Correct OTP submitted	200 OK with username	PASS
POST /auth/verify-otp	Incorrect OTP submitted	401 Unauthorized	PASS
POST /chat	Valid shopping query	200 OK with products array	PASS
POST /chat	Empty query string	422 Unprocessable Entity	PASS
POST /chat	Query with no Firecrawl match	200 OK (fallback context used)	PASS

16.3 LLM Output Validation

- High-specificity queries: 'Sony WH-1000XM5 vs Bose QC45, which is better for commuting under 30,000 rupees?' — validated correct named product comparison.

- Low-specificity queries: 'I want to buy a gift for my friend' — validated proactive recommendations without requesting more information.
- Multi-constraint queries: 'laptop for college, lightweight, good battery, under 60k, not for gaming' — validated correct multi-criteria filtering.
- Follow-up queries (memory test): First query 'wireless earphones', second query 'anything with longer battery?' — validated FAISS memory correctly surfaced the past query.
- Preference extraction test: Verified that extracted_preferences correctly captured budget and category from each query.

16.4 Frontend Integration Testing

End-to-end frontend testing covered: complete sign-up and sign-in flow with both authentication methods; OTP request and verification; sending a product query and verifying response rendering; clicking 'View Details' links; creating multiple chats and switching between them; searching chats by title; deleting a chat; and logging out.

16.5 Validation Summary

Test Category	Total Cases	Passed	Pass Rate
API Endpoint Tests	18	18	100%
LLM Output Format Validation	25	24	96%
Frontend Workflow Tests	22	22	100%
Authentication Flow Tests	12	12	100%
Error Handling Tests	10	9	90%

CHAPTER 17: RESULTS AND DISCUSSION

17.1 System Output

ShopMate AI successfully delivers on its core promise: transforming natural language shopping queries into structured, actionable product recommendations. Across all tested product categories including consumer electronics, fashion, home appliances, fitness equipment, books, and gifts, the system consistently generated relevant, well-reasoned, and correctly formatted responses.

17.2 Performance Evaluation

Response latency was measured across 30 test queries under standard broadband conditions. The median end-to-end response time (from query submission to product cards rendered in the browser) was 3.8 seconds, with a 95th percentile of 6.2 seconds.

Component	Median Latency	95th Percentile	Notes
FAISS search + embedding	< 150 ms	< 200 ms	CPU-based, very consistent
Firecrawl API call	1.4 s	2.8 s	Varies with query complexity
Groq API call	1.9 s	3.8 s	Varies with output length
FastAPI processing overhead	< 50 ms	< 100 ms	Negligible
Frontend render time	< 100 ms	< 200 ms	Negligible
End-to-end (total)	3.8 s	6.2 s	Within 5s target for most queries

17.3 AI Quality Assessment

The quality of AI-generated recommendations was evaluated subjectively across 50 test queries. Relevance was rated as 'good' or 'excellent' in 92% of test cases. Price estimates were expressed as ranges rather than precise prices, accurately reflecting LLM knowledge uncertainty. Reasoning quality was uniformly high, with the model providing specific, coherent justifications for each recommendation. Product diversity was maintained in 88% of cases.

CHAPTER 18: ADVANTAGES AND LIMITATIONS

18.1 Advantages

18.1.1 Multi-Source Intelligence Fusion

ShopMate AI's most significant architectural advantage is the fusion of three complementary intelligence sources: the LLM's broad world knowledge, Firecrawl's real-time web product data, and the FAISS vector memory's within-session preference continuity. This fusion architecture is the key differentiator from both pure LLM chatbots (which lack current data) and pure search tools (which lack reasoning capability).

18.1.2 Natural Language Interface

The conversational interface eliminates the friction associated with complex filter-based search UIs. Users can express multi-dimensional preferences in a single natural language message without needing to understand technical product specifications or navigate nested filter menus.

18.1.3 Structured, Actionable Output

Unlike general-purpose AI assistants that provide recommendations as unstructured text, ShopMate AI renders recommendations as structured cards with prices, reasons, and purchase links. This output format is directly actionable and scannable, enabling quick comparison across the recommended products.

18.1.4 Zero-Cost Deployable Architecture

The system is designed to run entirely on free-tier cloud resources (Render for backend, Vercel for frontend, Groq free tier for LLM inference, Firecrawl free tier for web search), making it accessible for academic demonstration without infrastructure investment.

18.1.5 Dual Authentication Flexibility

The combination of password and OTP authentication accommodates diverse user preferences and security requirements. The seamless toggle between methods in the same UI screen improves the authentication user experience significantly.

18.2 Limitations

18.2.1 In-Memory Vector Store Volatility

The FAISS index is maintained in process memory and is lost whenever the backend server restarts. This means cross-session preference memory is unavailable. For Render's free tier, which restarts services after inactivity periods, this limitation is frequently encountered.

18.2.2 Plaintext Credential Storage

User passwords are stored in plaintext in users.json. This is a significant security vulnerability that would need to be addressed with bcrypt password hashing before any real-user deployment.

18.2.3 Rate Limits on Free API Tiers

Both the Groq API and the Firecrawl API enforce rate limits on their free tiers. Heavy usage by multiple concurrent users would quickly exhaust these limits, causing API failures surfaced as error messages in the chat interface.

18.2.4 No Verified Purchase Links

The purchase URLs generated by the LLM are Amazon search links constructed from the product name rather than verified links to specific product listings. This means users are directed to search results rather than specific product pages.

18.2.5 No Actual Price Data

Price ranges provided by the system are estimates derived from the LLM's training data and Firecrawl search results, not real-time data from product databases. A production version would need to integrate with a price comparison API for accurate pricing.

CHAPTER 19: FUTURE ENHANCEMENTS

19.1 Persistent Vector Memory (Pinecone / pgvector)

Migrating the FAISS in-memory index to a persistent vector database such as Pinecone, Weaviate, or Supabase's pgvector extension would enable preference memory that persists across sessions and server restarts. The VectorStore interface is already abstracted in db/vector_store.py, making the migration straightforward.

19.2 Real-Time Price API Integration

Integrating with a product price API such as the Amazon Product Advertising API or a price comparison service would enable accurate, current prices and verified purchase links, transforming ShopMate AI from a recommendation suggestion tool to a genuine price-finding utility.

19.3 Secure Password Hashing

Implementing bcrypt or Argon2 password hashing for the users.json credential store, along with migration to a proper database backend such as PostgreSQL or MongoDB, would address the current security vulnerability and enable real-user deployment.

19.4 Personalization Engine with Long-Term Preference Learning

A dedicated personalization engine maintaining a structured user preference profile (preferred brands, typical budget ranges, product categories of interest) would enable significantly improved recommendation quality for returning users. The profile could be updated after each session using the extracted_preferences field already returned by the LLM.

19.5 Multi-Modal Product Search (Image Input)

Adding image upload capability to the chat input would enable users to submit photos of products they've seen and want to find alternatives for. This capability would require integration with a multi-modal LLM that can interpret visual content alongside text.

19.6 Mobile Application (React Native)

Developing a React Native mobile application would make ShopMate AI accessible to the large and growing mobile shopping audience. The existing FastAPI backend is fully mobile-compatible — only the frontend would need to be adapted.

19.7 Product Comparison Mode

A dedicated product comparison feature that accepts two or more product names and generates a structured side-by-side comparison table of specifications, pros, cons, and pricing would complement the recommendation capability with a research-mode tool.

19.8 Wishlist and Cart Integration

Adding a persistent wishlist feature enabling users to save recommended products for later review, share wishlists, and receive notifications about price drops would transform ShopMate AI from a single-session tool to an ongoing shopping companion.

CHAPTER 20: CONCLUSION

ShopMate AI demonstrates that the combination of modern large language model APIs, real-time web intelligence, and vector-based semantic memory can produce a genuinely useful, production-quality conversational shopping assistant. The system successfully addresses the core problem of inadequate product discovery in e-commerce by enabling users to describe their needs in natural language and receive curated, well-reasoned, and immediately actionable product recommendations.

The technical architecture of ShopMate AI represents a carefully designed integration of five distinct technologies — FastAPI, Groq Llama-3.3-70B, Firecrawl, FAISS, and React — each chosen for its specific contribution to the system's capabilities. The three-stage recommendation pipeline (vector memory retrieval, web trend fetch, LLM synthesis) is an architectural pattern that is broadly applicable to AI-augmented information retrieval systems beyond the specific use case of shopping assistance.

The prompt engineering work that drives the system's AI behavior represents a significant portion of the intellectual contribution of this project. The design of a system prompt that reliably produces correctly structured JSON output, prevents conversational anti-patterns, enables progressive preference extraction, and maintains appropriate diversity across recommendations required careful iteration and testing. The resulting prompt serves as a practical template for structured LLM output generation in similar applications.

The full-stack implementation demonstrates mastery of a comprehensive set of modern software development skills: asynchronous Python web development with FastAPI, multi-service AI API integration, vector embedding and similarity search, React state management

for complex conversational UIs, Tailwind CSS utility-class design system, dual-mode authentication implementation, and cloud deployment configuration.

ShopMate AI has several known limitations in its current form — most notably the in-memory vector store, plaintext credential storage, and estimated rather than real-time pricing. These limitations are acknowledged and form the basis of a clear enhancement roadmap. The architecture has been deliberately designed to make each of these limitations addressable through targeted component replacements without requiring systemic redesign.

In summary, ShopMate AI is a successful proof of concept for AI-driven product recommendation, a practically deployable web application, and an exemplary demonstration of full-stack AI engineering skills acquired through the B.Tech program at SRM University AP.

APPENDIX

A. Complete Source Code Overview

A.1 Project Folder Structure

```
GENAI PROJECT/
├── backend/
│   ├── db/
│   │   ├── users.json              # User credential store
│   │   └── vector_store.py         # FAISS vector memory module
│   ├── services/
│   │   ├── llm_service.py          # Groq LLM recommendation
service
│   │   └── firecrawl_service.py    # Firecrawl web search service
│   ├── main.py                     # FastAPI application entry
point
│   └── .env                        # Environment variables (not in
repo)
```

```
├── frontend/
│   ├── src/
│   │   ├── components/
│   │   │   ├── ChatInput.jsx
│   │   │   ├── ChatWindow.jsx
│   │   │   ├── MessageBubble.jsx
│   │   │   └── ProductCard.jsx
│   │   ├── services/api.js
│   │   ├── App.jsx
│   │   ├── App.css
│   │   └── main.jsx
│   ├── package.json
│   ├── tailwind.config.js
│   └── vite.config.js
├── theme_switcher.py
└── revert_theme.py
```

B. API Documentation

Method	Endpoint	Request Body	Response
POST	/auth/signup	{ email, password }	{ message, username }
POST	/auth/signin	{ email, password }	{ message, username }
POST	/auth/send-otp	{ email }	{ message } or { message, otp, type }
POST	/auth/verify-otp	{ email, otp }	{ message, username }
POST	/chat	{ query, preferences? }	{ message, products[], extracted_preferences }

C. System Prompt (Full Text)

- Persona: "You are ShopMate AI, an intelligent conversational shopping assistant."
- Memory constraint: "NEVER ask the user the same type of clarifying question repeatedly. If query lacks context, provide 3 diverse recommendations proactively."
- Output format: "You must respond strictly in JSON format" with the complete schema.

- URL fallback: "If a direct link is not known, generate an Amazon search link: https://www.amazon.com/s?k=[product+name]"
- Preference extraction: "Extract budget, category, and other preferences from the query and return them in extracted_preferences."

D. Configuration Files

D.1 tailwind.config.js

Registers three custom CSS animations: fade-in (opacity 0 to 1 over 0.5s), slide-up (opacity 0 + translateY(20px) to opacity 1 + translateY(0) over 0.4s), and pulse-slow (standard pulse at 3s interval).

D.2 vite.config.js

Configures Vite with the @vitejs/plugin-react plugin for JSX transformation and React Fast Refresh support. No custom build options are required beyond the plugin configuration.

D.3 .gitignore

The frontend .gitignore excludes node_modules, dist (build output), .env (environment variables), and common IDE-specific files from version control.

E. GitHub Repository

`https://github.com/sahithya624/Habit-Tracker.git`

The repository includes all source code files, configuration templates, and setup instructions in the README.md file. Sensitive credentials (API keys, SMTP passwords) are not included in the repository and must be configured separately in a .env file.

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— End of Report —